

Customer Support Ticket Analysis – Summary

Objective

The objective of this project was to analyze customer support ticket data using Python in Google Colab. The analysis focuses on managing ticket information, cleaning issue descriptions, extracting meaningful insights, and identifying patterns in customer complaints. The program stores ticket data such as ticket number, customer name, issue description, and priority level, and performs several analyses to understand the support workload and customer issues.

Dataset

The dataset initially contained **10 customer support tickets**, and additional tickets were added during execution, resulting in a total of **13 tickets analyzed**. Each ticket includes:

- Ticket Number
- Customer Name
- Issue Description
- Priority Level (High, Medium, Low)

The issue descriptions were cleaned by removing punctuation and unnecessary spaces to improve text analysis accuracy

Analysis Performed

1. Priority Distribution Analysis

The project analyzed the number of tickets in each priority category to understand the urgency of customer issues.

Findings:

Priority Level	Number of Tickets
High Priority	5
Medium Priority	4
Low Priority	4

This shows that **most issues fall under High priority with 5 tickets overall**, while a smaller portion require immediate attention.

2. Longest Issue Description

The program identified the ticket with the **longest issue description based on word count**.

Findings:

- **Ticket Number:** 2
- **Word Count:** 5 words

This ticket contains the most detailed customer complaint in the dataset.

3. Unique Word Analysis

The program combined all issue descriptions and extracted **unique words** by converting the list of words into a set.

Finding:

- **Total Unique Words:** 37

This analysis helps identify the variety of terms customers use when describing their problems and can help detect common issue patterns.

Key Insights

- The majority of tickets fall under **medium and low priority**, indicating that many issues are manageable without immediate escalation.
- Only **Five tickets are high priority**, which require urgent attention from support teams.
- The longest issue description contained **five words**, suggesting that most customer complaints in this dataset are brief.
- The dataset contains **37 unique words**, reflecting a moderate diversity in how customers describe their issues.

Conclusion

This project demonstrates how Python can be used for **basic customer support data analysis and text processing**. By analyzing ticket priorities, issue descriptions, and word patterns, organizations can better understand customer concerns and improve support service efficiency. Such analysis helps prioritize urgent issues and identify recurring problems in customer support systems.